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INSTITUTE OF ENGINEERING AND TECHNOLOGY
Approved by AICTE & Affiliated to APJ Abdul Kalam Technological University
(Owned by Adi Sankara Trust)
DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE)

AIL 411-DEEP LEARNING LAB

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AP,CSE(AI),ASIET,KALADY

EXPERIMENT 1

Familiarize basic python packages for deep learning such as Keras, Tensorflow etc.

Agenda

- Python
  - Introduction
  - Installation
  - Virtual Environments
  - IDEs
- Deep Learning Libraries
  - TensorFlow
  - PyTorch
  - Keras
- Practical Session
  - Classification using MNIST dataset

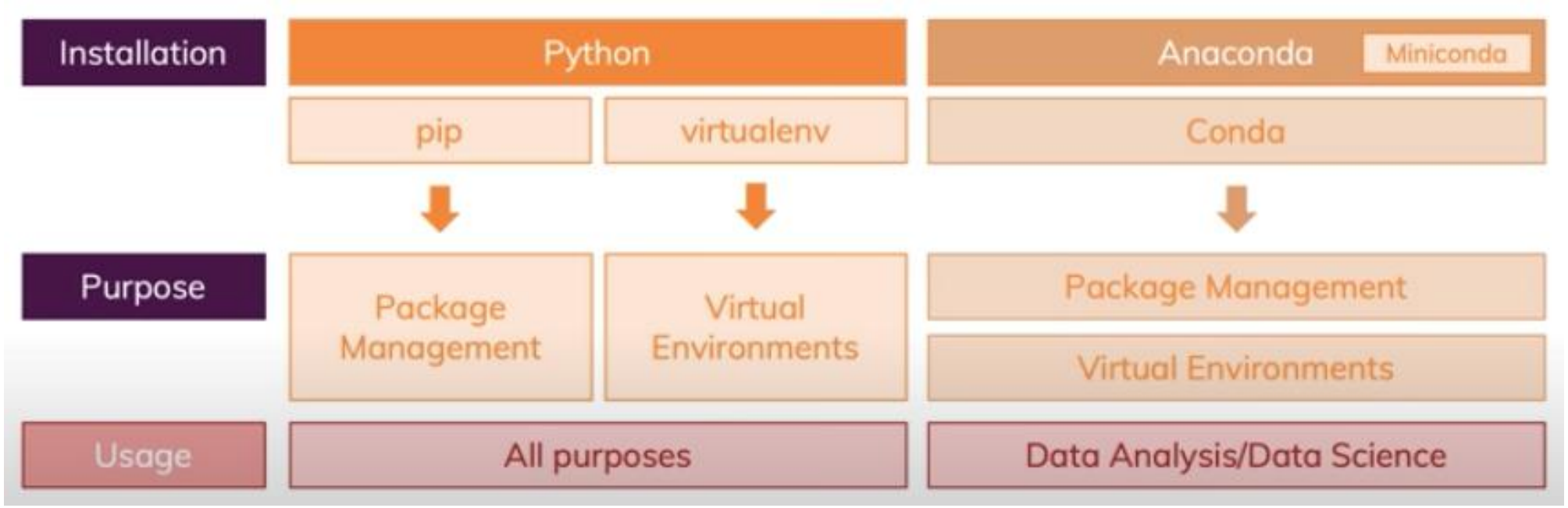
Python

- **Interpreted**: processed at run time
- **Interactive**: use a prompt and interacts with interpreter directly
- **Object Oriented**: supports OO Programming
- **Beginners Language**: Simple to Learn and Use
- **Extensive libraries**: Strong Support for ML and DL using different libraries

Anaconda distribution

- Python Distribution: Anaconda includes the Python programming language and its standard libraries.
- Conda Package Manager: Anaconda uses the conda package manager, which facilitates easy package installation and version management. (pip in python)
- Data Science Libraries: Anaconda comes pre-packaged with popular data science libraries like Pandas, NumPy, SciPy, Matplotlib, and scikit-learn.
- Virtual Environments: Anaconda allows users to create isolated environments to manage dependencies and package versions for different projects.

Python vs Anaconda

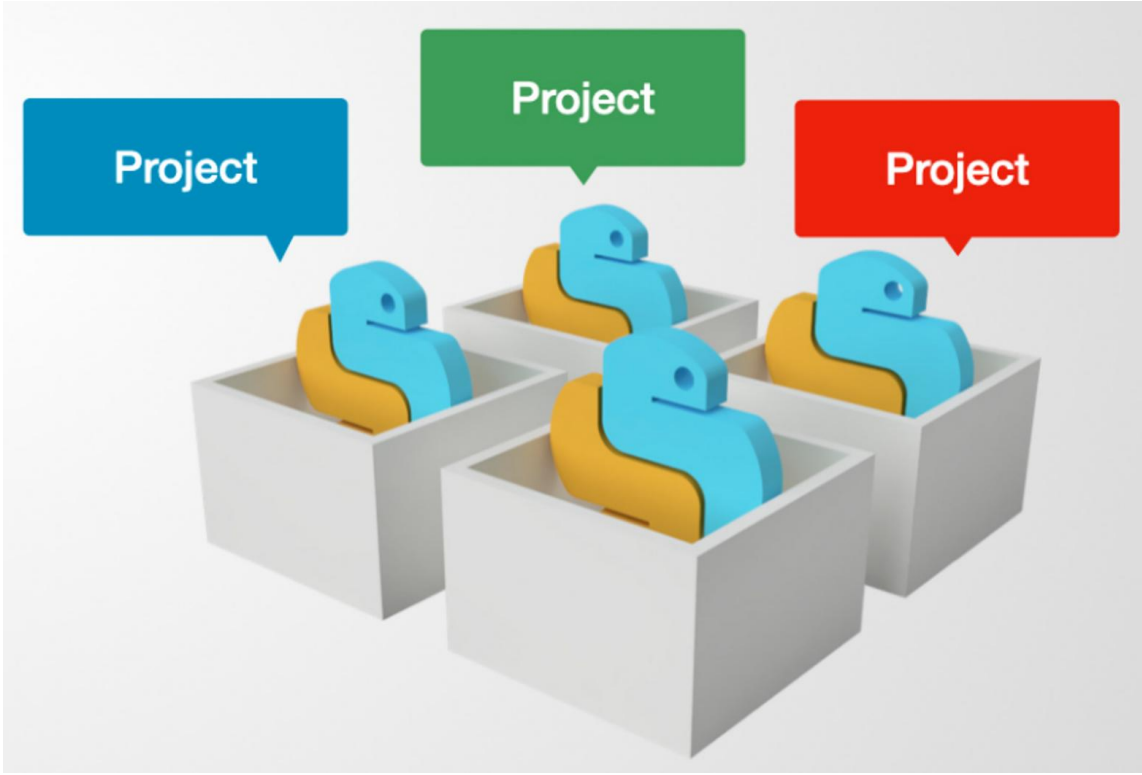


How to install Anaconda

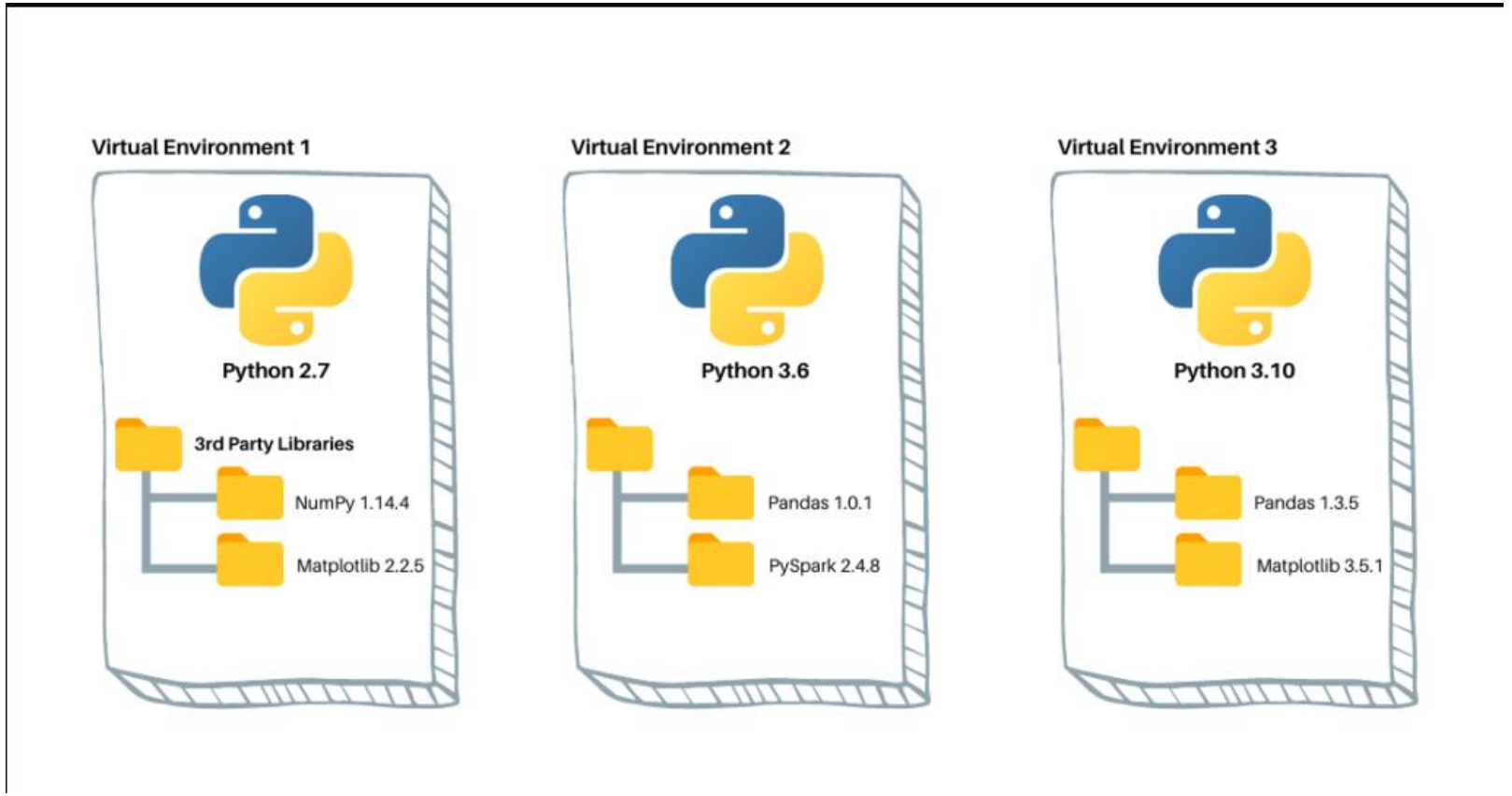
https://www.anaconda.com/download#download

Python Virtual Environments

- **Isolates** Python interpreter, libraries and scripts used in a specific project
- Enables **multiple** Environments in your computer, each with different versions of Python and its packages without interfering with each other



Python Virtual Environments



Conda Commands -Managing environments

https://anaconda.cloud/conda-cheatsheet

```
# Create an environment for technology with conda create -n {env}
conda create --name my_python_env

# Clone existing environment with conda create --clone {old_env} --name {new_env}
conda create --clone template_env --name project_env

# Create environment, auto-accepting prompts with conda create --yes --name {env}
# For non-interactive usage
conda create --yes --name my_env

# Make environment the default environment with conda activate {env}
# This prepends the environment directory to the system PATH environment variable
conda activate my_env

# Make the base environment the default with conda deactivate {env}
# This removes the environment directory from system PATH environment variable
conda deactivate my_env
```

Create Env-pratical session

1. Create an environment – name = dlLab
2. enable dlLab environment
3. list all the installed packages in the dlLab environment

Python Packages

Packages are collections of pre-written code and functions that extend the capabilities of the Python programming language. This can be reused instead of creating a new one for every project.

Data Exploration and Transformation

- NumPy
- pandas
- Jupyter
- Intake
- Dask

Visualization

- Matplotlib
- seaborn
- Plotly
- Bokeh
- HoloViews

AI and Machine Learning

- scikit-learn
- TensorFlow
- Keras
- PyTorch
- XGBoost

Natural Language Processing

- NLTK
- Gensim
- Transformers
- spaCy

Python Packages

- Matplotlib
  - This library is responsible for plotting numerical data. And that's why it is used in data analysis. It is also an open-source library and plots high-defined figures like pie charts, histograms, scatterplots, graphs, etc.
- Pandas
  - Pandas are an important library for data scientists. It is an open-source machine learning library that provides flexible high-level data structures and a variety of analysis tools. It eases data analysis, data manipulation, and cleaning of data. Pandas support operations like Sorting, Re-indexing, Iteration, Concatenation, Conversion of data, Visualizations, Aggregations, etc.
- Numpy
  - The name "Numpy" stands for "Numerical Python". It is the commonly used library. It is a popular machine learning library that supports large matrices and multi-dimensional data. It consists of in-built mathematical functions for easy computations. Even libraries like TensorFlow use Numpy internally to perform several operations on tensors. Array Interface is one of the key features of this library.
- SciPy
  - The name "SciPy" stands for "Scientific Python". It is an open-source library used for high-level scientific computations. This library is built over an extension of Numpy. It works with Numpy to handle complex computations. While Numpy allows sorting and indexing of array data, the numerical data code is stored in SciPy. It is also widely used by application developers and engineers.
- Scikit-learn
  - It is a famous Python library to work with complex data. Scikit-learn is an open-source library that supports machine learning. It supports variously supervised and unsupervised algorithms like linear regression, classification, clustering, etc. This library works in association with Numpy and SciPy.

Conda Commands - Installing and Managing Packages

```
# Install packages with conda install {pkg1} {pkg2} ...
conda install numpy pandas

# Install specific version of package with conda install {pkg}={version}
conda install scipy=1.10.1

# Update all packages with conda update --all
conda update --all

# Uninstall packages with conda uninstall {pkg}
conda uninstall pycaret
```

Install packages-pratical

Install

Python IDEs





- **PyCharm** is a powerful and feature-rich IDE developed by JetBrains. Preferred IDE for professional Python development
- **VS Code** :- is a lightweight, yet powerful, code editor developed by Microsoft
- **Jupyter** :- python notebook and web-based IDE that allows to execute code , view data and visual outputs, and write text all at one place.
  - Jupyter runs on your local machine
- **Colab** is also a notebook and are web-based IDE
  - Colab's runs on a Google server

Deep Learning Libraries



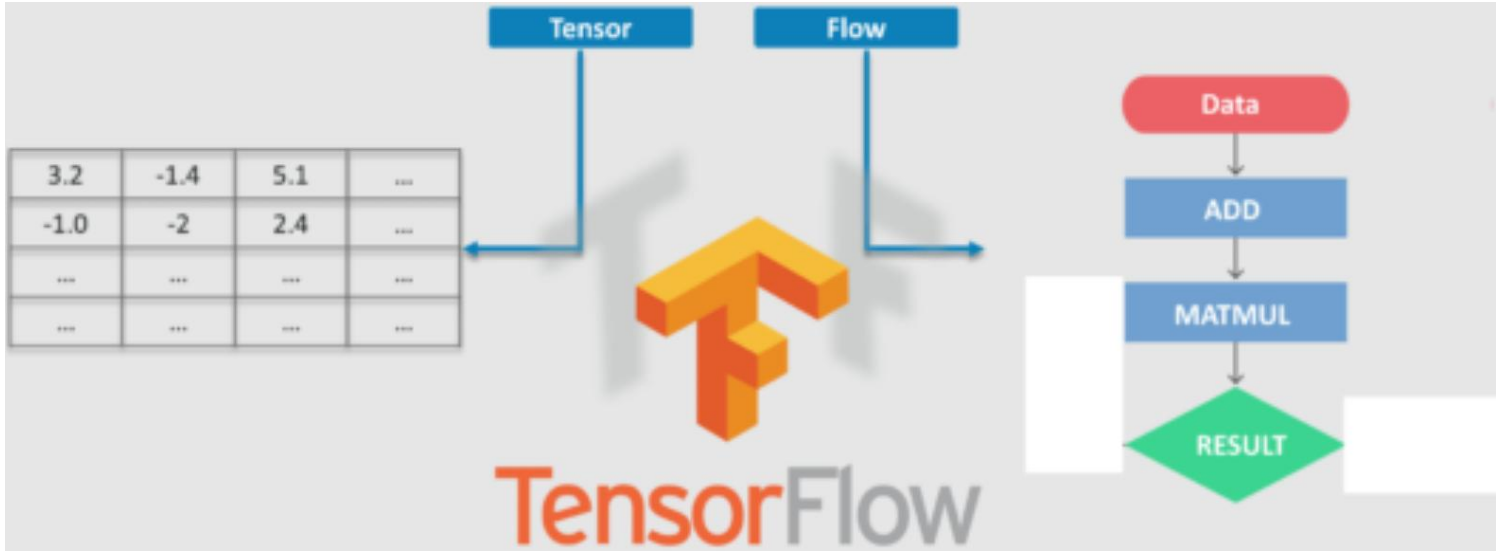
TensorFlow

- **TensorFlow** an open-source deep learning framework that offers a vast array of tools, libraries, and resources to create and implement machine learning and deep learning models.
- **TensorFlow** is and maintained developed by the Google Brain team
- Launched in 2015, TensorFlow is one of the best-known deep learning frameworks widely used by data science professionals.
- Currently, the stable version of TensorFlow is made open-source and is available under Apache License 2.0.

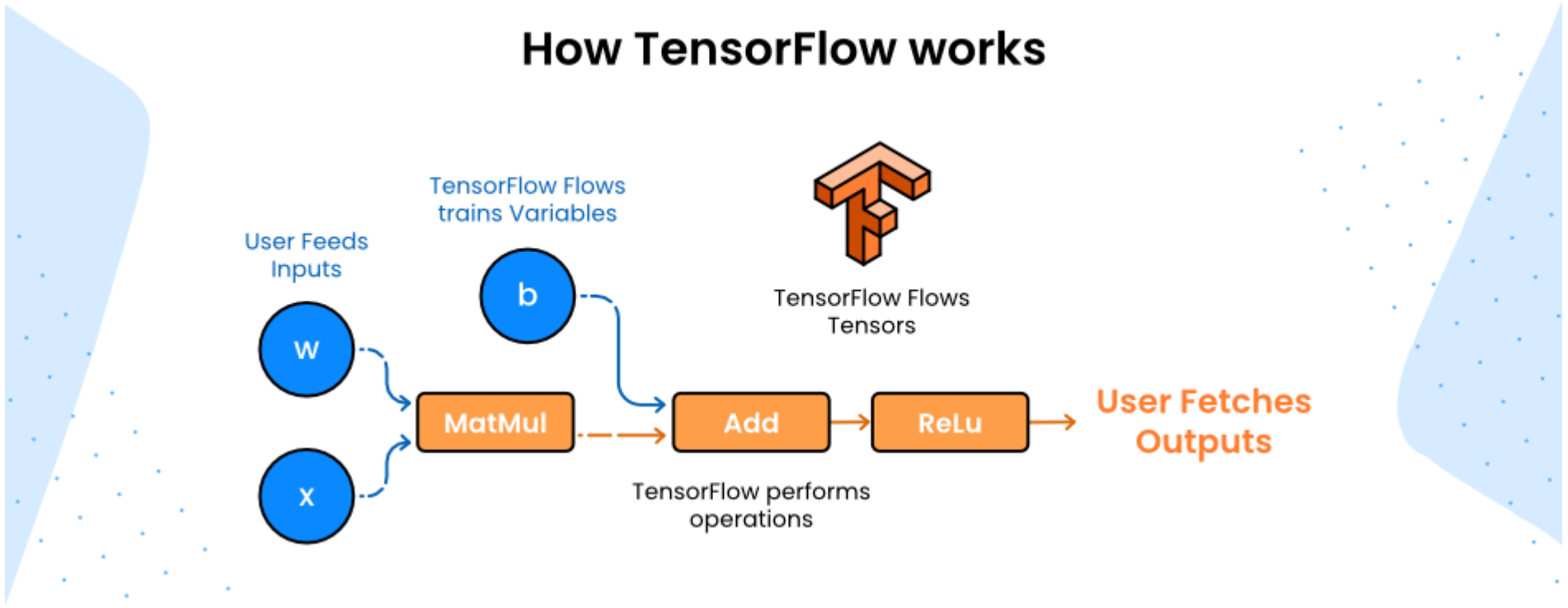
```
import tensorflow as tf
```

Tensors- Flow

- **TensorFlow** uses **tensors** to perform operations and manipulations in a neural network. A **tensor** is a multidimensional data array that flows through the neural net layers.



How TensorFlow works



PyTorch

PyTorch is Facebook's open-source machine learning framework. It was popularised in research in academia but increasingly has been used in production models.

- It follows graph architecture to define the logic of the neural net. That brings more clarity to the times of debugging and testing phase in the code.
- PyTorch allows users to create systems for computer vision, AI, personalization, and many more dynamics of deep learning.
- Because of the graph architecture it offers, PyTorch garnered more supporters and admiration among the developers compared to TensorFlow.

JAX

- Jax is a Python library designed for high-performance ML research
- JAX and TensorFlow were both written by Google. From my initial experimentation, JAX seems much easier to develop in and is more intuitive.
- JAX lacks the extensive infrastructure that TensorFlow has built over the years - be it open source projects, pre-trained models, tutorials, higher level abstractions (via Keras)
- JAX is a new machine learning framework that has been gaining popularity in machine learning research.

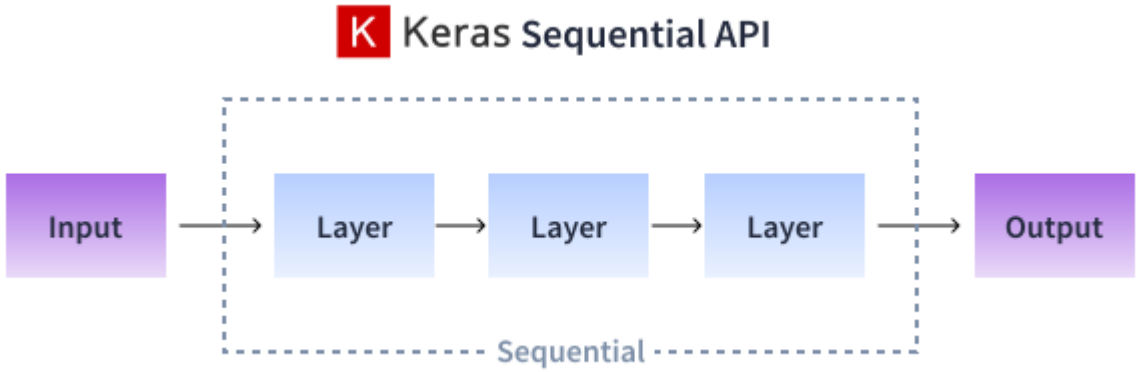
Introduction to Keras

- Keras is a high-level, deep-learning API developed by François Chollet, a Google engineer for implementing neural networks on 2015
- It is written in Python and is used to simplify the implementation of neural networks.
- It also supports multiple backend neural network computations.



Keras

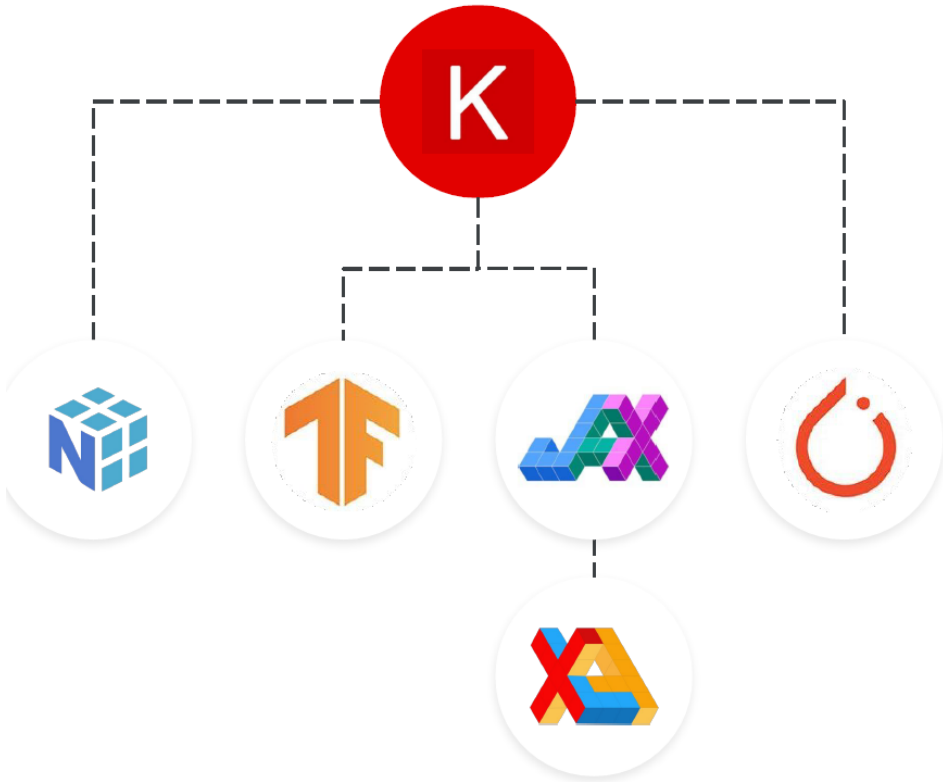
Uses Sequential API, neural network layers are organized in a linear, sequential manner, with data flowing through them in a single directio




```
model = keras.Sequential()
model.add(layers.Dense(16, activation="relu", name="layer1"))
model.add(layers.Dense(8, activation="relu", name="layer2"))
model.add(layers.Dense(4, name="layer3"))
```

Keras – New development – Keras 3

- For the past 4 years, Keras has exclusively supported the TensorFlow backend
- From Keras 3.0 onwards, Keras will be multi-backend again!
- Support for TensorFlow, JAX, PyTorch,NumPy backends

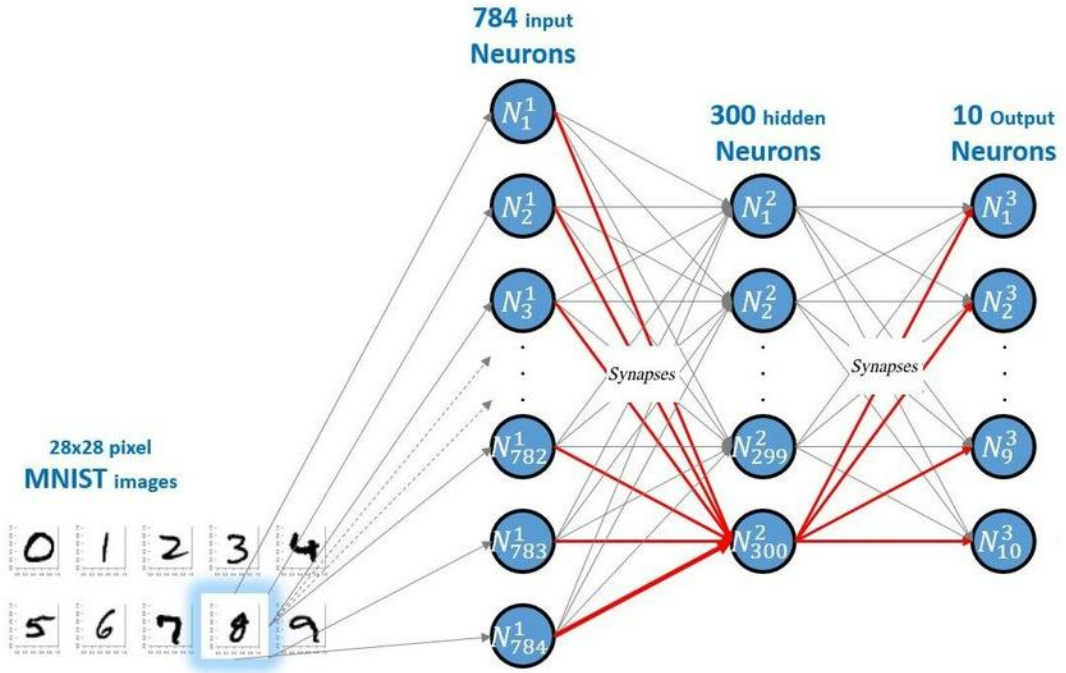


Keras vs TensorFlow vs PyTorch

Features	TensorFlow	PyTorch	Keras
Written In	C++, CUDA, Python	Lua	Python
Architecture	Not easy to use	Complex, less readable	Simple, concise, readable
API Level	High and Low	Low	High
Datasets	Large datasets, high-performance	Large datasets, high-performance	Smaller datasets
Debugging	Difficult to conduct debugging	Good debugging capabilities	Simple network, so debugging is not often needed
Does It Have Trained Models?	Yes	Yes	Yes
Popularity	Second most popular	Third most popular	Most Popular
Speed	Fast, high-performance	Fast, high-performance	Slow, low performance

PRACTICAL SESSION

MNIST




```
import tensorflow as tf
mnist = tf.keras.datasets.mnist

(x_train, y_train),(x_test, y_test) =
mnist.load_data()
x_train, x_test = x_train / 255.0, x_test / 255.0

model = tf.keras.models.Sequential([
  tf.keras.layers.Flatten(input_shape=(28, 28)),
  tf.keras.layers.Dense(300, activation='relu'),
  tf.keras.layers.Dense(10, activation='softmax')
])

model.compile(optimizer='adam',
  loss='sparse_categorical_crossentropy',
  metrics=['accuracy'])

model.fit(x_train, y_train, epochs=5)
model.evaluate(x_test, y_test)
```

THANK YOU